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COURSE CODE: SBT-DF204
LAB NAME: INVESTIGATING DJI MAVIC AIR FLIGHT EVIDENCE
DATE: 02/1012026
INSTRUCTOR: AMINU IDRIS

This individual case study asks you to examine DJI Mavic Air evidence from an Android mobile extraction, a raw internal microSD image, and an Expert Witness Format image. You will identify and compare DJI GO 4 flight records, cached video, detailed records, cached pictures, and media artefacts. Your report must build an evidence-led flight timeline while clearly separating direct observations from inferences about a flight, location, device, or operator.

Case scenario

Investigators recovered an Android logical acquisition associated with the DJI GO 4 application and two representations of DJI Mavic Air internal microSD media. The enquiry is whether the sources support a reliable account of aircraft activity, recorded media, and the relationship between the mobile application and microSD evidence. The case material may contain flight locations, images, and records relating to people. This is an offline examination of authorised course artefacts. It does not authorise you to fly, power on, connect to, synchronise, reset, or change settings on any drone, mobile device, account, cloud service, or removable media.

Case materials

Use only the Android logical image and report, the raw Mavic internal microSD image, the EWF image, and accompanying teaching decks supplied through E-Campus. The DJI GO 4 application package is dji.go.v4. The mobile deck introduces four artefact families: main flight records in TXT format, cached MP4 video, detailed DAT flight records, and cached pictures. The microSD decks cover raw and EWF examination. Use the exact filenames supplied in your package, create working copies, and record the source and integrity value of each original before analysis.

What you must submit

Item	Required output	What to provide
1	Forensic working set	Record the Android logical image, mobile forensic report, raw microSD image, and EWF image by original file name, byte size, SHA-256 value, source location, and working-copy location. Preserve every original unchanged.
2	DJI GO 4 evidence record	Provide labelled screenshots or saved output for the dji.go.v4 artefacts: main flight records, cached video, detailed flight records, cached pictures, associated metadata, and your time-zone treatment.

3	Media comparison record	Document the raw microSD and EWF image formats, partitions or file listings, relevant media hashes and timestamps, microSD identifier or image serial information, and cross-source comparison.
4	Student report	Submit one report in PDF or DOCX format with a completed evidence log and a UTC flight timeline. Include only the supporting files requested by your lecturer.

Investigation task

Conduct the investigation in stages: preserve the supplied sources; inventory DJI GO 4 artefacts; interpret each record type according to its event scope; examine the raw microSD image; examine the EWF image; compare the material sources; then create one UTC timeline. A main flight record, video cache, and detailed record use different start and end conditions, so you must not assume that their timestamps describe the same event. Base each finding on identifiable data such as a file path, hash, parser output, media identifier, image serial, timestamp, metadata field, or screenshot.

No	Investigation question	Minimum evidence to discuss
1	How did you preserve the case materials?	For each mobile and microSD source, record the source, original file name, byte size, SHA-256 value, working-copy method, tool version, and read-only or access precautions.
2	What DJI GO 4 artefacts are present?	Locate and document the DJI GO 4 application data and identify the folders or paths used for main flight records, cached video, detailed flight records, and cached pictures.
3	What does the main flight record establish?	Use the supplied course workflow to interpret the relevant TXT record. Identify device or aircraft information, start and end events, location evidence, duration, source time, UTC conversion, and any time-zone anomaly.
4	What does the cached video establish?	Identify the MP4 source path, filename, start and end information, duration, metadata, hash, and its relationship to the main flight record. Preserve metadata before previewing a working copy.
5	What does the detailed flight record establish?	Examine the DAT record using the approved course workflow. Explain its power-on to power-off scope and how it corroborates or differs from motor-on, take-off, landing, and video-recording evidence.
6	What do cached pictures establish?	Record paths, file types, capture sequence, timestamps, hashes, and the limits of using visual content or cache artefacts to infer a flight, location, or operator.
7	What does the raw internal microSD image show?	Document image format, integrity, sector or partition layout, read-only mount method, relevant videos and pictures, file metadata, hashes, and the microSD identifier if available.
8	What does the EWF image show and how does it compare?	Verify format and integrity, examine the mounted EWF data, record the image serial or media identifier, and compare media hashes, timestamps, and other artefacts with the raw image.

9	What conclusion can you defend?	Build a UTC flight timeline, distinguish direct observations from inference, state a confidence level, and explain at least three limitations or alternative explanations about device ownership, operator identity, route, and time.

Technical starting points

The following are course-aligned starting points. They do not replace documentation of your own method. Work only on copies, keep the original sources unchanged, and preserve the output that supports each decision.

Purpose	Course-aligned starting point
Preserve and identify evidence	Create read-only originals and working copies. Record SHA-256 values before and after copying; record the supplied mobile report as a reference source, not as a substitute for your own verification.
Locate DJI GO 4 artefacts	Use the Android logical extraction and the course-provided mobile report to locate dji.go.v4 data. Record full paths for TXT, MP4, DAT, and cached-picture artefacts.
Interpret main flight records	Use the approved offline parser or manual workflow demonstrated in class. Record the input file, tool or parser, version, source time, UTC conversion, and any ambiguity in the displayed time zone.
Validate video and pictures	Record file type, byte size, filename, timestamps, and SHA-256 value before opening a working copy. Use metadata output and a controlled preview, then preserve the supporting screenshot.
Examine raw microSD media	Use a read-only mount or forensic tool workflow. Document image type, sector or partition structure, file listings, timestamps, hashes, and relevant media identifiers.
Examine EWF media	Use the approved EWF workflow, such as ewfmount <image> <mount_point>, then inspect the exposed data read-only. Record the image format, integrity result, serial or GUID information, and access permissions.
Corroborate sources	Compare hashes, timestamps, file names, media identifiers, record scopes, and content only after normalising time. Explain both matches and differences between the mobile, raw microSD, and EWF sources.

Evidence interpretation safeguards

- Treat each record type according to its stated scope. A record beginning at power on, motor on, take-off, video start, landing, or power off supports a different part of the sequence.
- Document the source time zone, UTC conversion, and any apparent device time-setting problem. Do not use a time in a filename as proof without validating it against record content and other artefacts.
- A file hash can establish binary equality between copies, but it does not alone prove when a flight occurred, who operated the aircraft, or where it was flown.
- Use a read-only workflow for the microSD images. Record any mount location, permission or ownership change, tool version, partition view, and output path so another examiner can repeat your steps.
- Handle cached pictures, video, GPS information, and records about people as sensitive evidence. Redact unrelated personal or location data in screenshots while retaining enough forensic detail for assessment.
- Do not upload case files to external drone portals, flight parsers, cloud accounts, public scanners, or media-sharing services. Do not connect the evidence to live DJI equipment or execute any untrusted artefact.

Required report structure

1. Case reference and student identification. Include your full name, registration number, course code, and case title.

2. Executive conclusion. State a concise, cautious conclusion and confidence level. Do not present a person-level attribution unless the evidence supports it.
3. Evidence acquisition and integrity. Record sources, filenames, sizes, hashes, originals, working copies, tools, versions, and access controls.
4. Method. Explain the mobile-artifact, media-metadata, raw microSD, EWF, comparison, and time-normalisation methods used.
5. Findings. Answer every investigation question with a precise evidence reference: path, hash, parser output, timestamp, media identifier, image serial, metadata field, or screenshot ID.
6. Timeline and correlation. Present the material sequence in UTC and explain how TXT, MP4, DAT, cached pictures, raw microSD, and EWF evidence corroborate or conflict.
7. Conclusion and limitations. Explain gaps, altered access times, cache artefact limits, time-zone uncertainty, partial media, app versus device ownership, and the difference between technical records and proof of a person at the controls.
8. Appendices. Include labelled screenshots, selected tool output, the completed timeline, completed evidence log, and only the supporting material requested by your lecturer.

Timeline template

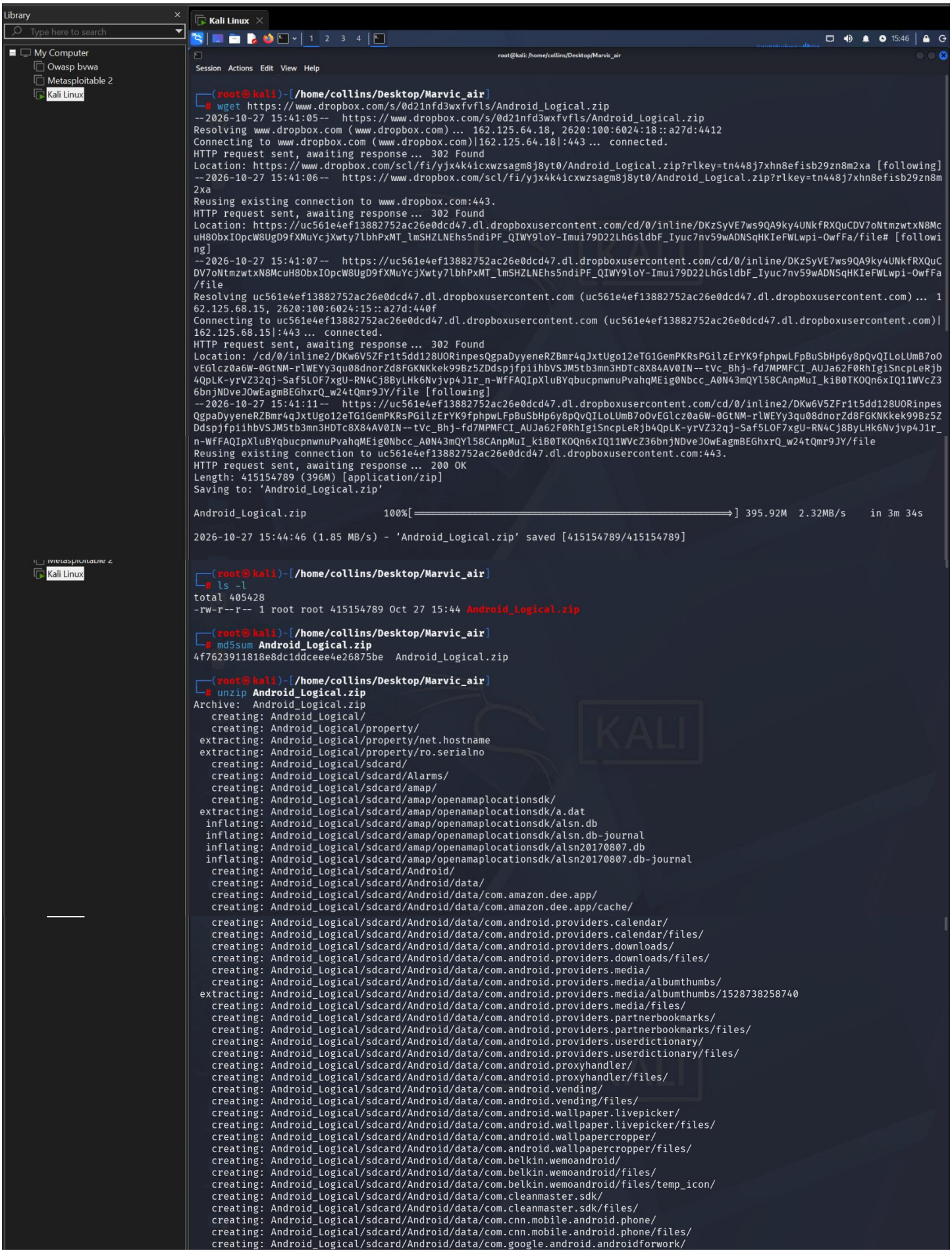
Complete a version of this table in your report or evidence appendix. Add rows for all material events and keep the main timeline in UTC.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Device state and power or motor on					
Take off or video start					
Main flight record event					
Landing or video end					
Power off and correlation					

Evidence log template

Use an evidence log that lets an assessor locate the exact source file, path, parser, mount method, hash, timestamp, media identifier, image serial, and screenshot that supports each finding.

Evidence ID	Tool and method	Source record, path, or image	Observed fact	Screenshot or appendix reference
E01				
E02				
E03				
E04				
E05				
E06				
E07				
E08				


Library
Type here to search
My Computer
Owasp bvwa
Metasploitable 2
Kali Linux
Kali Linux
1 2 3 4
15:46
root@kali: /home/collins/Desktop/Marvic_air
Session Actions Edit View Help
┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# wget https://www.dropbox.com/s/0d21nfd3wxfvfls/Android_Logical.zip
--2026-10-27 15:41:05-- https://www.dropbox.com/s/0d21nfd3wxfvfls/Android_Logical.zip
Resolving www.dropbox.com (www.dropbox.com)... 162.125.64.18, 2620:100:6024:18::a27d:4412
Connecting to www.dropbox.com (www.dropbox.com)|162.125.64.18|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/scl/fi/yjx4k4icxwzsagm8j8yt0/Android_Logical.zip?rlkey=tn448j7xhn8efisb29zn8m2xa [following]
--2026-10-27 15:41:06-- https://www.dropbox.com/scl/fi/yjx4k4icxwzsagm8j8yt0/Android_Logical.zip?rlkey=tn448j7xhn8efisb29zn8m2xa
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://uc561e4ef13882752ac26e0dcd47.dl.dropboxusercontent.com/cd/0/inline/DKzSyVE7ws9QA9ky4UNkfRXQuCDV7oNtmzwtxN8McuH8ObxIOpcW8UgD9fXMuYcjXwty7lbhPxMT_lmSHZLNEhs5ndiPF_QIWY9loY-Imui79D22LhGsldbF_Iyuc7nv59wADNSqHKIeFWLwpi-OwfFa/file# [following]
--2026-10-27 15:41:07-- https://uc561e4ef13882752ac26e0dcd47.dl.dropboxusercontent.com/cd/0/inline/DKzSyVE7ws9QA9ky4UNkfRXQuCDV7oNtmzwtxN8McuH8ObxIOpcW8UgD9fXMuYcjXwty7lbhPxMT_lmSHZLNEhs5ndiPF_QIWY9loY-Imui79D22LhGsldbF_Iyuc7nv59wADNSqHKIeFWLwpi-OwfFa/file
Resolving uc561e4ef13882752ac26e0dcd47.dl.dropboxusercontent.com (uc561e4ef13882752ac26e0dcd47.dl.dropboxusercontent.com)... 162.125.68.15, 2620:100:6024:15::a27d:440f
Connecting to uc561e4ef13882752ac26e0dcd47.dl.dropboxusercontent.com (uc561e4ef13882752ac26e0dcd47.dl.dropboxusercontent.com)|162.125.68.15|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DKw6V5ZFr1t5dd128UORinpesQgpaDyyeneRZBmr4qJxtUgo12eTG1GemPKRsPGilzErYK9fphpwLFpBuSbHp6y8pQvQILoLUmB7oOvEGlcz0a6W-0GtNM-rlWEYy3qu08dnorZd8FGKNKkek99Bz5ZDdspjfpiihbVSJM5tb3mn3HDTc8X84AV0IN--tVc_Bhj-fd7MPMFCI_AUJa62F0RhIgiSncpLeRjb4QpLK-yrVZ32qj-Saf5LOF7xgU-RN4Cj8ByLHk6Nvjvp4J1r_n-WfFAQIpXluBYqbucpnwnuPvahqMEig0Nbcc_A0N43mQYl58CAnpMuI_kiB0TKOQn6xIQ11WVcZ36bnjNDveJOwEagmBEGhxrQ_w24tQmr9JY/file [following]
--2026-10-27 15:41:11-- https://uc561e4ef13882752ac26e0dcd47.dl.dropboxusercontent.com/cd/0/inline2/DKw6V5ZFr1t5dd128UORinpesQgpaDyyeneRZBmr4qJxtUgo12eTG1GemPKRsPGilzErYK9fphpwLFpBuSbHp6y8pQvQILoLUmB7oOvEGlcz0a6W-0GtNM-rlWEYy3qu08dnorZd8FGKNKkek99Bz5ZDdspjfpiihbVSJM5tb3mn3HDTc8X84AV0IN--tVc_Bhj-fd7MPMFCI_AUJa62F0RhIgiSncpLeRjb4QpLK-yrVZ32qj-Saf5LOF7xgU-RN4Cj8ByLHk6Nvjvp4J1r_n-WfFAQIpXluBYqbucpnwnuPvahqMEig0Nbcc_A0N43mQYl58CAnpMuI_kiB0TKOQn6xIQ11WVcZ36bnjNDveJOwEagmBEGhxrQ_w24tQmr9JY/file
Reusing existing connection to uc561e4ef13882752ac26e0dcd47.dl.dropboxusercontent.com:443.
HTTP request sent, awaiting response... 200 OK
Length: 415154789 (396M) [application/zip]
Saving to: 'Android_Logical.zip'
Android_Logical.zip 100%[==========>] 395.92M 2.32MB/s in 3m 34s
2026-10-27 15:44:46 (1.85 MB/s) - 'Android_Logical.zip' saved [415154789/415154789]
┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# ls -l
total 405428
-rw-r--r-- 1 root root 415154789 Oct 27 15:44 Android_Logical.zip
┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# md5sum Android_Logical.zip
4f7623911818e8dc1ddceee4e26875be Android_Logical.zip
┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# unzip Android_Logical.zip
Archive: Android_Logical.zip
creating: Android_Logical/
creating: Android_Logical/property/
extracting: Android_Logical/property/net.hostname
extracting: Android_Logical/property/ro.serialno
creating: Android_Logical/sdcard/
creating: Android_Logical/sdcard/Alarms/
creating: Android_Logical/sdcard/amap/
creating: Android_Logical/sdcard/amap/openamaplocationsdk/
extracting: Android_Logical/sdcard/amap/openamaplocationsdk/a.dat
inflating: Android_Logical/sdcard/amap/openamaplocationsdk/alsn.db
inflating: Android_Logical/sdcard/amap/openamaplocationsdk/alsn.db-journal
inflating: Android_Logical/sdcard/amap/openamaplocationsdk/alsn20170807.db
inflating: Android_Logical/sdcard/amap/openamaplocationsdk/alsn20170807.db-journal
creating: Android_Logical/sdcard/Android/
creating: Android_Logical/sdcard/Android/data/
creating: Android_Logical/sdcard/Android/data/com.amazon.dee.app/
creating: Android_Logical/sdcard/Android/data/com.amazon.dee.app/cache/
creating: Android_Logical/sdcard/Android/data/com.android.providers.calendar/
creating: Android_Logical/sdcard/Android/data/com.android.providers.calendar/files/
creating: Android_Logical/sdcard/Android/data/com.android.providers.downloads/
creating: Android_Logical/sdcard/Android/data/com.android.providers.downloads/files/
creating: Android_Logical/sdcard/Android/data/com.android.providers.media/
creating: Android_Logical/sdcard/Android/data/com.android.providers.media/albumthumbs/
extracting: Android_Logical/sdcard/Android/data/com.android.providers.media/albumthumbs/1528738258740
creating: Android_Logical/sdcard/Android/data/com.android.providers.media/files/
creating: Android_Logical/sdcard/Android/data/com.android.providers.partnerbookmarks/
creating: Android_Logical/sdcard/Android/data/com.android.providers.partnerbookmarks/files/
creating: Android_Logical/sdcard/Android/data/com.android.providers.userdictionary/
creating: Android_Logical/sdcard/Android/data/com.android.providers.userdictionary/files/
creating: Android_Logical/sdcard/Android/data/com.android.proxyhandler/
creating: Android_Logical/sdcard/Android/data/com.android.proxyhandler/files/
creating: Android_Logical/sdcard/Android/data/com.android.vending/
creating: Android_Logical/sdcard/Android/data/com.android.vending/files/
creating: Android_Logical/sdcard/Android/data/com.android.wallpaper.livepicker/
creating: Android_Logical/sdcard/Android/data/com.android.wallpaper.livepicker/files/
creating: Android_Logical/sdcard/Android/data/com.android.wallpapercropper/
creating: Android_Logical/sdcard/Android/data/com.android.wallpapercropper/files/
creating: Android_Logical/sdcard/Android/data/com.belkin.wemoandroid/
creating: Android_Logical/sdcard/Android/data/com.belkin.wemoandroid/files/
creating: Android_Logical/sdcard/Android/data/com.belkin.wemoandroid/files/temp_icon/
creating: Android_Logical/sdcard/Android/data/com.cleanmaster.sdk/
creating: Android_Logical/sdcard/Android/data/com.cleanmaster.sdk/files/
creating: Android_Logical/sdcard/Android/data/com.cnn.mobile.android.phone/
creating: Android_Logical/sdcard/Android/data/com.cnn.mobile.android.phone/files/
creating: Android_Logical/sdcard/Android/data/com.google.android.androidforwork/

```
   creating: Android_Logical/sdcard/DJI/assistant/
   creating: Android_Logical/sdcard/DJI/dji.go.v4/
   creating: Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_33177fb665068406_1523611394000.jpg
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_5be64acbef28816c_1523610990000.jpg
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_6d9f39f0ca9decc3_1523610988000.jpg
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_7f92b744bbc53b3a_1523610992000.jpg
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_dd609e61934788b3_1523611494000.jpg
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_ed8c1a297e1061b4_1524152656000.jpg
   creating: Android_Logical/sdcard/DJI/dji.go.v4/databases/
   creating: Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.info
 extracting: Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
   creating: Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/analytics/
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/analytics/2018_06_19_14_50_39.map
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/analytics/2018_06_19_14_50_39.tlv
   creating: Android_Logical/sdcard/DJI/dji.go.v4/DJI_SPALSH/
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/DJI_SPALSH/2018-05-09.jpg
   creating: Android_Logical/sdcard/DJI/dji.go.v4/editor/
   creating: Android_Logical/sdcard/DJI/dji.go.v4/editor/auto/
   creating: Android_Logical/sdcard/DJI/dji.go.v4/editor/music/
   creating: Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cfg/
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cfg/music.cfg
  inflating: Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cfg/music2.cfg
   creating: Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/
 extracting: Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/BEYOND_THE_HORIZON.png
 extracting: Android_Logical/sdcard/DJI/dji.go.v4/editor/music/cover/BREAK_IT.png
```

```
  inflating: Android_Logical/sdcard/DJI/dji.pilot/LOG/CACHE/VerPhone/log-2017-09-27.txt
  inflating: Android_Logical/sdcard/DJI/dji.pilot/LOG/CACHE/VerPhone/log-2017-10-09.txt
  inflating: Android_Logical/sdcard/DJI/dji.pilot/LOG/CACHE/VerPhone/log-2017-10-10.txt
   creating: Android_Logical/sdcard/DJI/dji.pilot/LOG/ERROR_POP_LOG/
  inflating: Android_Logical/sdcard/DJI/dji.pilot/LOG/ERROR_POP_LOG/10-10-2017
   creating: Android_Logical/sdcard/DJI/dji.pilot/Package/
   creating: Android_Logical/sdcard/DJI/dji.pilot/RECORD_VOICE/
   creating: Android_Logical/sdcard/DJI/dji.pilot/Upgrade/
   creating: Android_Logical/sdcard/DJI/dji.pilot/Upgrade/DLPackage/
   creating: Android_Logical/sdcard/DJI/dji.pilot/VideoEditor/
   creating: Android_Logical/sdcard/DJI/dji.pilot/VideoEditor/production/
   creating: Android_Logical/sdcard/DJI/dji.pilot/xtScreenshots/
   creating: Android_Logical/sdcard/Download/
   creating: Android_Logical/sdcard/Movies/
   creating: Android_Logical/sdcard/Music/
   creating: Android_Logical/sdcard/Notifications/
   creating: Android_Logical/sdcard/Pictures/
   creating: Android_Logical/sdcard/Playlists/
   creating: Android_Logical/sdcard/Podcasts/
   creating: Android_Logical/sdcard/Ringtones/
   creating: Android_Logical/sdcard/Samsung/
   creating: Android_Logical/sdcard/Samsung/Music/
  inflating: Android_Logical/sdcard/Samsung/Music/Over the Horizon.mp3
   creating: Android_Logical/sdcard/TWRP/

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# ls
Android_Logical  Android_Logical.zip

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# tree -L 1 Android_Logical/sdcard/DJI/dji.go.v4/
Android_Logical/sdcard/DJI/dji.go.v4/
├── CACHE_IMAGE
├── databases
├── DJI_RECORD
├── DJI_SPALSH
├── editor
├── FlightRecord
├── LOG
├── Package
├── RECORD_VOICE
├── VideoEditor
├── videoguide
├── visual_guide
└── xtScreenshots

14 directories, 0 files

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# tree Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord
Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord
├── DJIFlightRecord_2018-06-19_[14-50-34].txt
└── MCDatFlightRecords
    ├── 18-06-19-02-47-38_FLY057.DAT
    └── 18-06-19-02-59-05_FLY058.DAT

2 directories, 3 files
```

Metasploitable 2

Kali Linux

Metasploitable 2
Kali Linux

```
┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# stat Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
  File: Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
  Size: 555776          Blocks: 1088       IO Block: 4096   regular file
Device: 8,1     Inode: 3934436     Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-10-27 16:11:47.985111194 +0100
Modify: 2018-06-19 12:55:40.000000000 +0100
Change: 2026-10-27 15:50:36.601191389 +0100
 Birth: 2026-10-27 15:50:36.595126110 +0100

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# file Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt: data

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# xxd -s -200 Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/DJIFlightRecord_2018-06-19_[14-50-34].txt
00087a38: 3239 4596 bb46 2680 10f6 e274 c7ff ff64  29E..F&....t...d
00087a48: 726f 6e65 666f 2e2e 2e2d 4d61 7669 6320  ronefo...-Mavic
00087a58: 4169 7200 0000 0000 0000 0000 0000 0030  Air............0
00087a68: 4b31 4446 3331 3342 4434 5052 3100 0000  K1DF313BD4PR1...
00087a78: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00087a88: 0000 0000 0000 0000 0000 0000 0000 0030  ...............0
00087a98: 4b34 4145 4251 4133 3430 3044 5400 0002  K4AEBQA3400DT...
00087aa8: 0402 1000 0000 0000 0000 0000 0100 0000  ................
00087ab8: 0000 0000 0000 000f 3ed6 0d40 42ac 1967  ........>..@B..g
00087ac8: 3db7 0c41 44e6 124d 38f1 0e44 41a1 1a4e  =..AD..M8..DA..N
00087ad8: 3eb4 0540 42a6 194c 35b7 0d42 43a6 154c  >..@B..L5..BC..L
00087ae8: 38b5 0d41 4ea5 184f 38b6 0043 43a5 184c  8..AN..O8..CC..L
00087af8: 22b7 0c40 42a7 fcff                      "..@B...

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# tree Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord
Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord
├── DJIFlightRecord_2018-06-19_[14-50-34].txt
└── MCDatFlightRecords
    ├── 18-06-19-02-47-38_FLY057.DAT
    └── 18-06-19-02-59-05_FLY058.DAT

2 directories, 3 files

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# stat Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord
  File: Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord
  Size: 4096            Blocks: 8          IO Block: 4096   directory
Device: 8,1     Inode: 3934435     Links: 3
Access: (0755/drwxr-xr-x)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-10-27 16:06:57.016117291 +0100
Modify: 2018-10-03 11:57:48.000000000 +0100
Change: 2026-10-27 15:50:42.623753281 +0100
 Birth: 2026-10-27 15:50:36.594747732 +0100

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# tree -L 1 Android_Logical/sdcard/DJI/dji.go.v4/
Android_Logical/sdcard/DJI/dji.go.v4/
├── CACHE_IMAGE
├── databases
├── DJI_RECORD
├── DJI_SPALSH
├── editor
├── FlightRecord
├── LOG
├── Package
├── RECORD_VOICE
├── VideoEditor
├── videoguide
├── visual_guide
└── xtScreenshots

14 directories, 0 files

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# tree -L 2 Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/ --inodes
[3934271]  Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/
├── [3934272]  2018_06_19_14_50_39.info
├── [3934273]  2018_06_19_14_50_39.mp4
└── [3934274]  analytics
    ├── [3934275]  2018_06_19_14_50_39.map
    └── [3934276]  2018_06_19_14_50_39.tlv

2 directories, 4 files

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# tree -L 2 Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/ --inodes
[3934271]  Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/
├── [3934272]  2018_06_19_14_50_39.info
├── [3934273]  2018_06_19_14_50_39.mp4
└── [3934274]  analytics
    ├── [3934275]  2018_06_19_14_50_39.map
    └── [3934276]  2018_06_19_14_50_39.tlv

2 directories, 4 files

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# stat Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
  File: Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
  Size: 57791765        Blocks: 112888     IO Block: 4096   regular file
Device: 8,1     Inode: 3934273     Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-10-27 16:22:41.826685185 +0100
Modify: 2018-06-19 12:55:42.000000000 +0100
Change: 2026-10-27 15:50:35.533451104 +0100
 Birth: 2026-10-27 15:50:34.135158123 +0100
```

```
┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# sudo apt-get install mediainfo
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
mediainfo is already the newest version (26.05-1).
Solving dependencies ... Done
0 upgraded, 0 newly installed, 0 to remove and 287 not upgraded.

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# mediainfo Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
General
Complete name                            : Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
Format                                   : MPEG-4
Format profile                           : Base Media
Codec ID                                 : isom (isom/iso2/avc1/mp41)
File size                                : 55.1 MiB
Duration                                 : 4 min 0 s
Overall bit rate                         : 1 919 kb/s
Frame rate                               : 29.236 FPS
Writing application                      : Lavf56.15.102

Video
ID                                       : 1
Format                                   : AVC
Format/Info                              : Advanced Video Codec
Format profile                           : High@L3.1
Format settings                          : CABAC / 1 Ref Frames
Format settings, CABAC                   : Yes
Format settings, Reference frames        : 1 frame
Format settings, Slice count             : 8 slices per frame
Codec ID                                 : avc1
Codec ID/Info                            : Advanced Video Coding
Duration                                 : 4 min 0 s
Bit rate                                 : 1 917 kb/s
Width                                    : 1 280 pixels
Height                                   : 720 pixels
Display aspect ratio                     : 16:9
Frame rate                               : 29.236 FPS
Color space                              : YUV
Chroma subsampling                       : 4:2:0
Bit depth                                : 8 bits
Scan type                                : Progressive
Bits/(Pixel*Frame)                       : 0.071
Stream size                              : 55.1 MiB (100%)
Writing library                          : x264 core 142
Encoding settings                        : cabac=1 / ref=1 / deblock=1:0:0 / analyse=0:0x111 / me=hex / subme=7 / psy=1 / psy_
rd=1.00:0.00 / mixed_ref=0 / me_range=16 / chroma_me=1 / trellis=0 / 8x8dct=0 / cqm=0 / deadzone=21,11 / fast_pskip=1 / chroma
_qp_offset=-2 / threads=1 / lookahead_threads=1 / sliced_threads=0 / slice_max_mbs=480 / slice_min_mbs=80 / nr=0 / decimate=1
/ interlaced=0 / bluray_compat=0 / constrained_intra=0 / bframes=0 / weightp=0 / keyint=1 / keyint_min=1 / scenecut=40 / intra
_refresh=0 / rc=crf / mbtree=0 / crf=26.0 / qcomp=0.60 / qpmin=26 / qpmax=26 / qpstep=4 / ip_ratio=1.40 / aq=0
Codec configuration box                  : avcC

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# sudo apt install libimage-exiftool-perl
libimage-exiftool-perl is already the newest version (13.55+dfsg-1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 287

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# exiftool Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD/2018_06_19_14_50_39.mp4
ExifTool Version Number         : 13.55
File Name                       : 2018_06_19_14_50_39.mp4
Directory                       : Android_Logical/sdcard/DJI/dji.go.v4/DJI_RECORD
File Size                       : 58 MB
File Modification Date/Time     : 2018:06:19 12:55:42+01:00
File Access Date/Time           : 2026:10:27 16:22:41+01:00
File Inode Change Date/Time     : 2026:10:27 15:50:35+01:00
File Permissions                : -rw-r--r--
File Type                       : MP4
File Type Extension             : mp4
MIME Type                       : video/mp4
Major Brand                     : MP4 Base Media v1 [IS0 14496-12:2003]
Minor Version                   : 0.2.0
Compatible Brands               : isom, iso2, avc1, mp41
Media Data Size                 : 57756891
Media Data Offset               : 48
Movie Header Version            : 0
Create Date                     : 0000:00:00 00:00:00
Modify Date                     : 0000:00:00 00:00:00
Time Scale                      : 1000
```

```
Duration                        : 0:04:01
Preferred Rate                  : 1
Preferred Volume                : 100.00%
Preview Time                    : 0 s
Preview Duration                : 0 s
Poster Time                     : 0 s
Selection Time                  : 0 s
Selection Duration              : 0 s
Current Time                    : 0 s
Next Track ID                   : 2
Track Header Version            : 0
Track Create Date               : 0000:00:00 00:00:00
Track Modify Date               : 0000:00:00 00:00:00
Track ID                        : 1
Track Duration                  : 0:04:01
Track Layer                     : 0
Track Volume                    : 0.00%
Matrix Structure                : 1 0 0 0 1 0 0 0 1
Image Width                     : 1280
Image Height                    : 720
Media Header Version            : 0
Media Create Date               : 0000:00:00 00:00:00
Media Modify Date               : 0000:00:00 00:00:00
Media Time Scale                : 90000
Media Duration                  : 0:04:01
Media Language Code             : und
Handler Description             : VideoHandler
Graphics Mode                   : srcCopy
Op Color                        : 0 0 0
Compressor ID                   : avc1
Source Image Width              : 1280
Source Image Height             : 720
X Resolution                    : 72
Y Resolution                    : 72
Bit Depth                       : 24
Video Frame Rate                : 29.236
Handler Type                    : Metadata
Handler Vendor ID               : Apple
Encoder                         : Lavf56.15.102
Image Size                      : 1280×720
Megapixels                      : 0.922
Avg Bitrate                     : 1.92 Mbps
Rotation                        : 0

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# tree Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/
Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/
├── DJIFlightRecord_2018-06-19_[14-50-34].txt
└── MCDatFlightRecords
    ├── 18-06-19-02-47-38_FLY057.DAT
    └── 18-06-19-02-59-05_FLY058.DAT

2 directories, 3 files

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# stat Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT
  File: Android_Logical/sdcard/DJI/dji.go.v4/FlightRecord/MCDatFlightRecords/18-06-19-02-47-38_FLY057.DAT
  Size: 4002791         Blocks: 7824       IO Block: 4096   regular file
Device: 8,1     Inode: 3934438     Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2018-06-19 12:58:24.000000000 +0100
Modify: 2018-06-19 12:58:24.000000000 +0100
Change: 2026-10-27 15:50:36.681171977 +0100
 Birth: 2026-10-27 15:50:36.606725532 +0100

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# tree -L 1 Android_Logical/sdcard/DJI/dji.go.v4/
Android_Logical/sdcard/DJI/dji.go.v4/
├── CACHE_IMAGE
├── databases
├── DJI_RECORD
├── DJI_SPALSH
├── editor
├── FlightRecord
├── LOG
├── Package
├── RECORD_VOICE
├── VideoEditor
├── videoguide
├── visual_guide
└── xtScreenshots
```

```
┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# tree Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE
Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE
├── thumb_33177fb665068406_1523611394000.jpg
├── thumb_5be64acbef28816c_1523610990000.jpg
├── thumb_6d9f39f0ca9decc3_1523610988000.jpg
├── thumb_7f92b744bbc53b3a_1523610992000.jpg
├── thumb_dd609e61934788b3_1523611494000.jpg
└── thumb_ed8c1a297e1061b4_1524152656000.jpg

1 directory, 6 files

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# ls --full-time -tr Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE
total 104
-rw-r--r-- 1 root root 19508 2018-06-19 12:47:50.000000000 +0100 thumb_7f92b744bbc53b3a_1523610992000.jpg
-rw-r--r-- 1 root root 12755 2018-06-19 12:47:50.000000000 +0100 thumb_6d9f39f0ca9decc3_1523610988000.jpg
-rw-r--r-- 1 root root 19191 2018-06-19 12:47:50.000000000 +0100 thumb_5be64acbef28816c_1523610990000.jpg
-rw-r--r-- 1 root root 15631 2018-06-19 12:47:52.000000000 +0100 thumb_ed8c1a297e1061b4_1524152656000.jpg
-rw-r--r-- 1 root root 13791 2018-06-19 12:47:52.000000000 +0100 thumb_dd609e61934788b3_1523611494000.jpg
-rw-r--r-- 1 root root 14224 2018-06-19 12:47:52.000000000 +0100 thumb_33177fb665068406_1523611394000.jpg

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# stat Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_33177fb665068406_1523611394000.jpg
  File: Android_Logical/sdcard/DJI/dji.go.v4/CACHE_IMAGE/thumb_33177fb665068406_1523611394000.jpg
  Size: 14224           Blocks: 32         IO Block: 4096   regular file
Device: 8,1     Inode: 3934264     Links: 1
Access: (0644/-rw-r--r--)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2018-06-19 12:47:52.000000000 +0100
Modify: 2018-06-19 12:47:52.000000000 +0100
Change: 2026-10-27 15:50:34.129160357 +0100
 Birth: 2026-10-27 15:50:34.129044886 +0100

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# ls --full-time -tr Android_Logical/sdcard/Android/data/dji.go.v4/cache
total 576
-rw-r--r-- 1 root root 139332 2018-06-19 11:02:20.000000000 +0100 component_upgrade_list.json
-rw-r--r-- 1 root root  31686 2018-06-19 11:02:20.000000000 +0100 component_upgrade_list_br.json
-rw-r--r-- 1 root root     30 2018-06-19 11:02:20.000000000 +0100 component_upgrade_date.json
-rw-r--r-- 1 root root      4 2018-06-19 11:13:22.000000000 +0100 afinalCache.1
-rw-r--r-- 1 root root 240032 2018-06-19 11:15:46.000000000 +0100 afinalCache.idx
-rw-r--r-- 1 root root   4643 2018-06-19 11:15:46.000000000 +0100 afinalCache.0
-rw-r--r-- 1 root root 139332 2018-06-19 11:36:24.000000000 +0100 list.json
-rw-r--r-- 1 root root     30 2018-06-19 12:12:56.000000000 +0100 date.json
drwxr-xr-x 2 root root   4096 2018-09-22 22:30:52.000000000 +0100 afinalCache
drwxr-xr-x 2 root root   4096 2018-10-03 11:57:24.000000000 +0100 uil-images
```

Kali Linux

```
┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# wget wget https://www.dropbox.com/s/6qghgnkhe7a8wga/df048_internal_microSD.001
Prepended http:// to 'wget'
--2026-11-02 10:21:15--  http://wget/
Resolving wget (wget)... failed: Name or service not known.
wget: unable to resolve host address ‘wget’
--2026-11-02 10:21:19--  https://www.dropbox.com/s/6qghgnkhe7a8wga/df048_internal_microSD.001
Resolving www.dropbox.com (www.dropbox.com)... 162.125.68.18, 2620:100:6023:18::a27d:4312
Connecting to www.dropbox.com (www.dropbox.com)|162.125.68.18|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/scl/fi/qjl1vb9sdfq3fafef6hus/df048_internal_microSD.001?rlkey=f0y1mz4wr5y73khh7fl0cz6ct [following]
--2026-11-02 10:21:38--  https://www.dropbox.com/scl/fi/qjl1vb9sdfq3fafef6hus/df048_internal_microSD.001?rlkey=f0y1mz4wr5y73khh7fl0cz6ct
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: ‘df048_internal_microSD.001’

df048_internal_microSD.001          [    <=>                                             ] 200.50K   289KB/s    in 0.7s

2026-11-02 10:21:43 (289 KB/s) - ‘df048_internal_microSD.001’ saved [205310]

FINISHED --2026-11-02 10:21:43--
Total wall clock time: 28s
Downloaded: 1 files, 200K in 0.7s (289 KB/s)
```

Kali Linux

```
┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# md5sum df048_internal_microSD.001
6a144cd23d0104f11e27fa425f4eb1dd  df048_internal_microSD.001

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# file df048_internal_microSD.001
df048_internal_microSD.001: HTML document, ASCII text, with very long lines (58016)

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# sudo losetup --partscan --find --show --read-only df048_internal_microSD.001
/dev/loop0
losetup: df048_internal_microSD.001: Warning: file does not end on a 512-byte sector boundary; the remaining end of the file will be ignored.

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# ls /media/kali/2019-1B01 -l
ls: cannot access '/media/kali/2019-1B01': No such file or directory

┌──(root㉿kali)-[/home/collins/Desktop/Marvic_air]
└─# lsblk -f
NAME   FSTYPE  FSVER            LABEL             UUID                                   FSAVAIL FSUSE% MOUNTPOINTS
loop0
sda
├─sda1 ext4    1.0                                7e4f5037-98c1-4049-961d-804a6efd90b7   44.3G    35% /
├─sda2
└─sda5 swap    1                                  c5dcff0b-1c47-4aae-9740-e39356cf0207                [SWAP]
sr0    iso9660 Joliet Extension Kali Linux amd64 1 2026-06-16-13-03-54-00
```